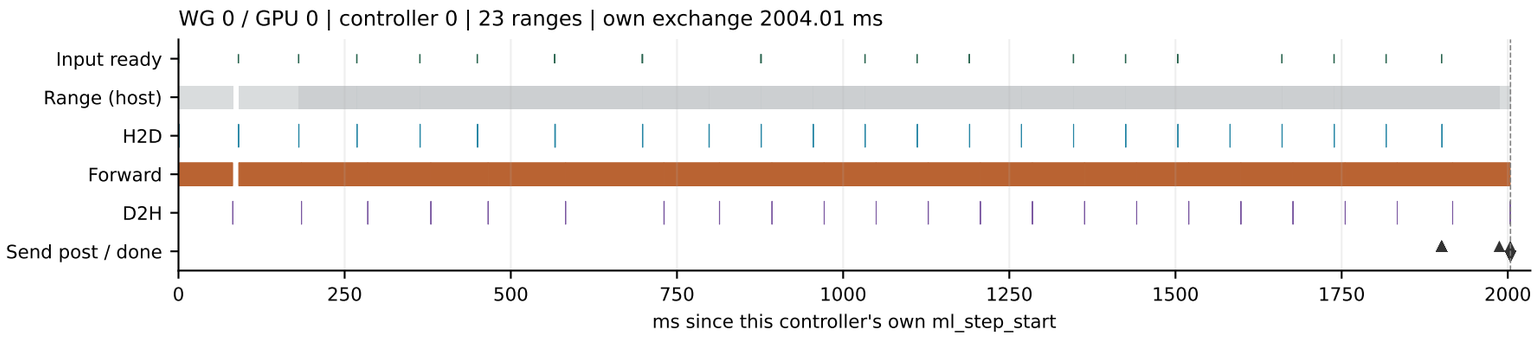
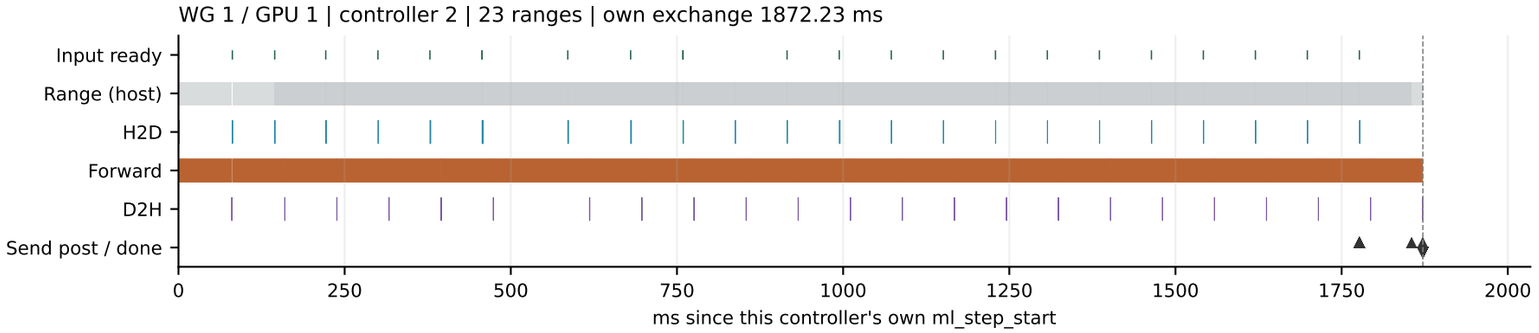
transformer | 96c4g | numa | collective vs p2p1 | rep 1 | call 6 / MAIA step 189

Overall MAIA ML-step makespan: collective 1693.84 ms | p2p1 2024.94 ms



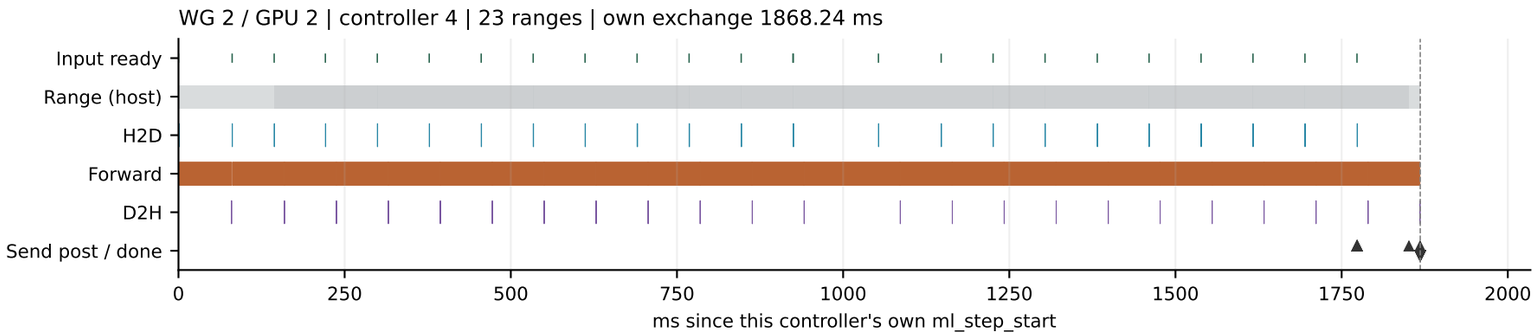
GPU work (ms)	Collective	p2p1
H2D	4.967	13.135
Forward	1615.280	1992.947
D2H	7.014	6.091
GPU SUM	1627.261	2012.174

Controllers: coll 0 / P2P 0
Coll host device elapsed: 1627.799 ms
P2P GPU envelope: 2003.619 ms
Different scopes; not an elapsed-time comparison.



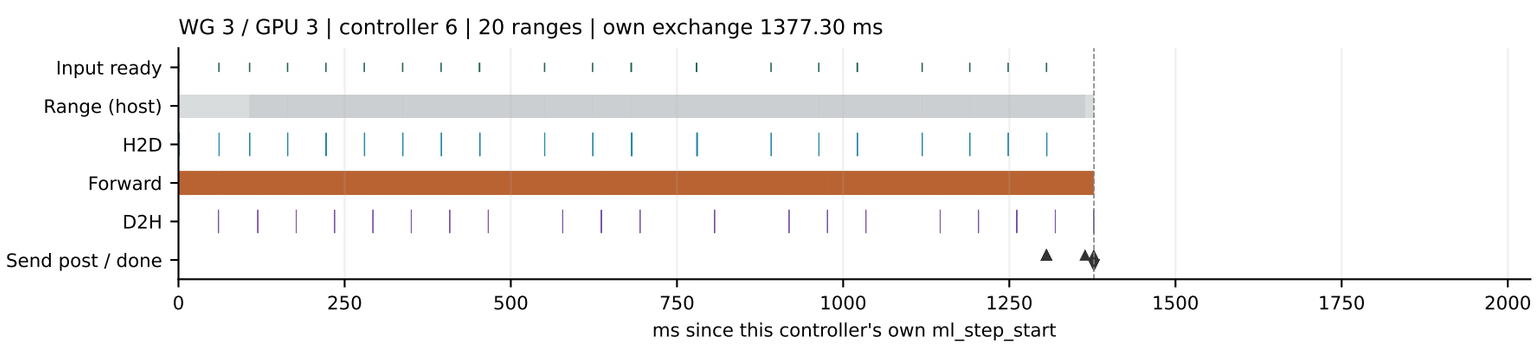
GPU work (ms)	Collective	p2p1
H2D	4.748	16.248
Forward	1616.324	1869.029
D2H	6.568	5.990
GPU SUM	1627.640	1891.267

Controllers: coll 2 / P2P 2
Coll host device elapsed: 1628.147 ms
P2P GPU envelope: 1871.840 ms
Different scopes; not an elapsed-time comparison.



GPU work (ms)	Collective	p2p1
H2D	4.957	10.976
Forward	1618.055	1864.894
D2H	3.915	4.716
GPU SUM	1626.928	1880.586

Controllers: coll 4 / P2P 4
Coll host device elapsed: 1627.424 ms
P2P GPU envelope: 1867.837 ms
Different scopes; not an elapsed-time comparison.



GPU work (ms)	Collective	p2p1
H2D	5.457	11.081
Forward	1231.899	1374.221
D2H	5.395	4.198
GPU SUM	1242.751	1389.500

Controllers: coll 6 / P2P 6
Coll host device elapsed: 1243.356 ms
P2P GPU envelope: 1376.803 ms
Different scopes; not an elapsed-time comparison.

Each controller has its own zero; rows are NOT globally synchronized. CUDA-event durations are reliable; host-anchored inter-request overlap is approximate. GPU work SUM is not elapsed time or a critical path.